📄 Titanic Dataset – Exploratory Data Analysis Report

🎯 Objective

To explore the Titanic dataset using Exploratory Data Analysis (EDA) techniques to uncover patterns, trends, and insights related to passenger survival.

📦 Dataset Overview

- **Rows**: 891 passengers
- **Columns**: 12 features
- **Target Variable**: Survived (0 = No, 1 = Yes)

🔍 Step 1: Data Exploration

- The dataset contains both numerical and categorical features.
- Missing values observed in Age, Cabin, and Embarked.
- The target variable (Survived) is imbalanced, with ~38% of the data indicating Survival.

📊 Step 2: Univariate Analysis

- **Sex**: There are more males (65%) than females.
- **Pclass**: Most passengers were in 3rd class.
- **Age**: Right-skewed; most were 20–40 years old.
- **Fare**: Right-skewed with outliers; most paid below $100.
- **Survival Rate**: Higher for females, younger passengers, and 1st class travelers.

🔗 Step 3: Bivariate Analysis

- **Sex vs Survived**: Females had much higher survival rates.
- **Pclass vs Survived**: 1st class passengers survived more.
- **Embarked vs Survived**: Highest survival from Cherbourg.
- **Age vs Survived**: Children under 10 had better survival odds.

- **Fare vs Survived**: Higher fares linked to higher survival.

🔄 Step 4: Multivariate Analysis

- **Pairplot** revealed strong survival clusters based on:
  - **Fare**: Higher fare → higher survival.
  - **Sex + Age**: Females and younger passengers more likely to survive.
- **Heatmap** showed:
  - Strong correlation between Survived and Sex, Pclass, Fare.
  - Weak correlation with SibSp, Parch.

🎨 Advanced Visuals

- **Violin Plot**: Female 1st class passengers had higher survival and wider age range.
- **Heatmap (pivot)**: Survival was highest for **1st class females**, lowest for **3rd class males**.

✅ Key Takeaways

- Survival was influenced by **sex, class, age**, and **fare**.
- **Females in 1st and 2nd class** had the best survival odds.
- **Males in 3rd class** had the lowest.
- **Fare and Class** played a major role in determining survival.

📌 Tools Used

- Python, Pandas, Matplotlib, Seaborn
- Google Colab for interactive coding and visualization